Amr Gomaa

Applied Machine Learning Researcher and PhD Candidate

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https://amrgomaaelhady.github.io/
https://umtl.cs.uni-saarland.de/people/amr-gomaa.html

Research Focus

I am broadly interested in research areas related to the intersection of **machine learning** (language and vision) with **human-computer interaction**. My research topics include **multimodal interaction**, **incremental learning**, and **reinforcement learning** in the Automotive and Robotics context. Additionally, I have worked on multiple industry-oriented interdisciplinary projects with industrial partners such as **BMW**, **IAV (Volkswagen Subsidiary)**, and **Zeiss**.

Education

Ph.D. in Computer Science *2020 - ongoing*

German Research Center for Artificial Intelligence (DFKI), Germany
Advisors: Dr.-Ing Michael Feld and Prof. Dr. Antonio Krüger

M.Sc. in Computer Science *2017 - 2020*

Saarland University, Germany
GPA: 1.4/1.0 (Thesis GPA: 1.3)
Advisor: Dr.-Ing Michael Feld and Prof. Dr. Antonio Krüger

B.Sc. in Telecommunications and Electronics Engineering *2008 - 2013*

Ain Shams University, Egypt

Miscellaneous

Udacity Deep Reinforcement Learning Nanodegree, 2023
Sixth Summer School on Computational Interaction (CIXSchool), 2022
First Inria-DFKI European Summer School on AI (IDESSAI), 2021

Research Experience

PhD Candidate *2020 - ongoing*

German Research Center for Artificial Intelligence (DFKI), Germany

Project Manager, TeachTAM Project *2022 - ongoing*

German Research Center for Artificial Intelligence (DFKI), Germany

Junior Researcher *2019 - 2020*

German Research Center for Artificial Intelligence (DFKI), Germany

Research Assistant *2017 - 2019*

Human-Computer Interaction Group, Saarland University, Germany
Advised by Prof. Dr. Jürgen Steimle

Industry Experience

Senior Radio Frequency (RF) Optimization Engineer *2016 - 2017*

Vodafone Egypt

Vodafone's Discover Graduate Program for Outstanding Academics *2015 - 2016*

Vodafone Egypt

Teaching Experience

Student Theses Mentor and Co-Superviser *2020 - ongoing*

- Kiran Gani, Master Thesis Title: Personalisation of Modalities using Reinforcement Learning for Object Referencing
- Simon Engel, Master Thesis Title: Adaptive User Interface Approach for Efficient Transfer of Control in Conditionally Automated Vehicles.
- Michael Sargious, Bachelor Thesis Title: An Adaptive Incremental End-to-End ML Framework with Automated Feature Engineering.
- Hassan Kanso, Bachelor Thesis Title: Multi-person Multi-camera tracking for Industry 4.0.

Lecturer & Teaching Assistant *2019 - ongoing*

Saarland University, Germany
Classes: Speech-based Adaptation of Personalized User Interfaces Seminar, AI for HCI Seminar, Adaptive Human Machine Interfaces for Autonomous Systems Seminar, Hybrid Machine Learning Approaches and Applications Seminar, Automotive User Interfaces Seminar, and Machine Learning School

Tutor *2017 - 2019*

Saarland University, Germany
Classes: Human Computer Interaction and Neural Networks: Implementation and Application

Publications

[1] **Amr Gomaa** and Michael Feld. "Towards Adaptive User-Centered Neuro-Symbolic Learning for Multimodal Interaction with Autonomous Systems". In: *AI & HCI Workshop at the 40th International Conference on Machine Learning (ICML) and Proceedings of the 2023 International Conference on Multimodal Interaction (ICMI)*. **Won Third Place for Blue Sky Papers**. 2023, pp. 689–694.

[2] Sahar Abdelnabi, **Amr Gomaa**, Sarath Sivaprasad, Lea Schönherr, and Mario Fritz. "LLM-Deliberation: Evaluating LLMs with Interactive Multi-Agent Negotiation Games". In: *arXiv preprint arXiv:2309.17234* (2023). In Submission.

[3] **Amr Gomaa**, Bilal Mahdy, Niko Kleer, Michael Feld, Frank Kirchner, and Antonio Krüger. "Teach Me How to Learn: A Perspective Review towards User-centered Neuro-symbolic Learning for Robotic Surgical Systems". In: *arXiv preprint arXiv:2307.03853* (2023). In Submission.

[4] **Amr Gomaa** and Bilal Mahdy. "Unveiling the Role of Expert Guidance: A Comparative Analysis of User-centered Imitation Learning and Traditional Reinforcement Learning". In: *The 1st International Workshop on Human-in-the-Loop Applied Machine Learning (HITLAML)*. 2023 (to appear), **Best Paper Award**.

[5] **Amr Gomaa**, Robin Zitt, and Guillermo Reyes. "SynthoGestures: A Novel Framework for Synthetic Dynamic Hand Gesture Generation for Driving Scenarios". In: *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*. 2023 (to appear).

[6] **Amr Gomaa**, Robin Zitt, and Guillermo Reyes. "Advancing Dynamic Hand Gesture Recognition in Driving Scenarios with Synthetic Data". In: *Adjunct Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI)*. 2023 (to appear).

[7] Guillermo Reyes*, **Amr Gomaa***, and Michael Feld. "It's all about you: Personalized in-Vehicle Gesture Recognition with a Time-of-Flight Camera". In: *Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI)*. *: Equal contribution. 2023 (to appear).

[8] **Amr Gomaa**, Alexandra Alles, Elena Meiser, Lydia Helene Rupp, Marco Molz, and Guillermo Reyes. "What's on your mind? A Mental and Perceptual Load Estimation Framework towards Adaptive In-vehicle Interaction while Driving". In: *Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI)*. 2022, pp. 215–225.

[9] Elena Meiser, Alexandra Alles, Samuel Selter, Marco Molz, **Amr Gomaa**, and Guillermo Reyes. "In-Vehicle Interface Adaptation to Environment-Induced Cognitive Workload". In: *Adjunct Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI)*. 2022, pp. 83–86.

[10] **Amr Gomaa**. "Adaptive User-Centered Multimodal Interaction towards Reliable and Trusted Automotive Interfaces". In: *Proceedings of the 2022 International Conference on Multimodal Interaction (ICMI)*. 2022, pp. 690–695.

[11] **Amr Gomaa**, Guillermo Reyes, and Michael Feld. "ML-PersRef: A Machine Learning-Based Personalized Multimodal Fusion Approach for Referencing Outside Objects From a Moving Vehicle". In: *Proceedings of the 2021 International Conference on Multimodal Interaction (ICMI)*. 2021, pp. 318–327.

[12] **Amr Gomaa**, Guillermo Reyes, Alexandra Alles, Lydia Rupp, and Michael Feld. "Studying person-specific pointing and gaze behavior for multimodal referencing of outside objects from a moving vehicle". In: *Proceedings of the 2020 International Conference on Multimodal Interaction (ICMI)*. 2020, pp. 501–509.

Academic Activities

Reviewing

AC (Associate Chair) member of AutomotiveUI WIP 2023, PC (Program Committee) member of HITLAML 2023 Workshop, IMWUT 2023, CHI 2023 (Special Recognition for Outstanding Review), IEEE VR 2023, NordiCHI 2022, AutomotiveUI (Special Recognition for Outstanding Review) 2023 & 2022 & 2021 , CHI PLAY 2021, ICMI 2021, and IEEE AIVR 2021.

Talks

Teach Me How to Learn: An Outlook on User-centered Neuro-Symbolic Learning for Robotic Surgical Systems
Medical & Environmental Computing (MEC-Lab) at TU Darmstadt, 2022

TeachTAM: Machine Teaching with Hybrid Neuro-symbolic Reinforcement Learning
Software Campus Summit, 2022

Awards

Won Third Place for Blue Sky Papers at ICMI'23, 2023
Research Grant (100,000€) from the German Federal Ministry of Education and Research (BMBF), 2022
Saarland University scholarship for international students (DAAD STIBET III scholarship grant), 2019

Languages

English (Bilingual)
German (Intermediate)
Colloquial Egyptian (Mother Tongue)

Referees

Prof. Dr. Antonio Krüger
CEO, German Research Center for Artificial Intelligence (DFKI). Professor and Media Informatics Chair, Saarland University
antonio.krueger@dfki.de

Dr. Michael Feld
Senior Researcher and Team Lead, German Research Center for Artificial Intelligence (DFKI)
michael.feld@dfki.de

Prof. Dr. Anna Maria Feit
Professor in Computer Science Department, Saarland University. Leading the Computational Interaction Group.
feit@cs.uni-saarland.de

Dr. Mridula Singh
Tenure-track Faculty at CISPA – Helmholtz Center for Information Security. Leading the Singh Group.
singh@cispa.de